

A Proposal for a Tiered Cognitive Architecture for AI Systems

Executive Summary

Current AI systems are built around a powerful computational engine: the large language model (LLM). As these systems have evolved, however, the language model has gradually become responsible for an expanding range of functions beyond probabilistic inference, including conversation state, memory management, constraint tracking, execution planning, tool routing, and logical verification.

This concentration of responsibility forces fundamentally different classes of computation through the same probabilistic engine. The result is structural inefficiency: repeated processing of conversational history, increasing computational cost, context degradation over long interactions, and growing cognitive friction for users forced into iterative debugging cycles.

This proposal explores a different architectural direction.

Rather than treating the language model as the operating system of an AI, it proposes separating executive cognition from computational inference. A persistent **Executive Cognitive Kernel** maintains system state, compiles task-specific execution contexts, and dispatches work to specialized computational engines.

The guiding principle is simple:

Different classes of cognitive work should be performed by the computational substrate best suited to perform them.

The Current Architecture and Its Structural Limitations

Modern AI systems consist of sophisticated infrastructure surrounding a language model, including CPUs, GPUs, retrieval systems, databases, serving engines, orchestration layers, and external tools.

Despite this complexity, the language model frequently serves as the de facto executive controller. Every interaction reconstructs conversational context, interprets goals, performs planning, and generates responses through repeated neural inference.

This architecture naturally produces several structural limitations.

Repeated Context Processing

Conversation history continually grows while much of its content becomes irrelevant to the current task. Although modern inference engines employ numerous optimizations, repeatedly reconstructing state from conversational history remains an architectural overhead independent of any specific transformer implementation.

Context Degradation

As conversations become longer, outdated assumptions, abandoned hypotheses, or earlier mistakes may continue influencing subsequent reasoning. Recovering from these errors often requires repeated corrective interactions.

Human Productivity Cost

When AI systems require multiple correction cycles before producing an acceptable result, the computational cost is only part of the problem. Users themselves become the debugging system, interrupting concentration, increasing cognitive load, and reducing overall productivity.

Architectural Principle

This proposal is not an incremental refinement of existing agentic systems.

It proposes relocating executive cognition outside the probabilistic inference engine, allowing specialized computational systems to perform the classes of reasoning for which they are naturally suited.

Language generation.

Logical reasoning.

Memory organization.

Planning.

Retrieval.

Optimization.

These are distinct computational problems.

They should not necessarily share the same computational mechanism.

Tier One — Executive Cognitive Kernel

The Executive Cognitive Kernel functions as the persistent operating system of the cognitive architecture.

Unlike today's stateless inference model, the Kernel maintains continuous cognitive state independent of any individual model invocation.

Its responsibilities include:

- Maintaining persistent cognitive state.
- Tracking goals, tasks, and user preferences.
- Managing structured knowledge.
- Selecting relevant information.
- Compiling task-specific execution contexts.
- Dispatching computation.
- Integrating returned results.
- Coordinating all specialized computational resources.

Importantly, the Executive Kernel is deterministic.

Its purpose is not to perform reasoning itself, but to coordinate reasoning.

Context Compilation

The Executive Kernel does not replay conversations.

It compiles context.

Rather than sending an entire transcript to a language model, the Kernel constructs a minimal execution payload derived from structured cognitive state.

This payload contains only the information necessary for the current task.

Temporary execution contexts are discarded after completion.

Persistent cognitive state remains independent of model inference.

This approach separates long-term knowledge from temporary working context while reducing the influence of obsolete conversational history.

Persistent Cognitive State

Rather than treating conversation transcripts as memory, the Executive Kernel maintains structured system state.

Examples include:

- user preferences
- established facts
- active projects
- current objectives
- hypotheses under evaluation

- procedural knowledge
- relationships between concepts

The Executive Kernel owns this state.

Language models consume selected portions of it.

Tier Two — Specialized Computation Layer

Tier Two contains heterogeneous computational resources optimized for different forms of reasoning.

The Executive Kernel determines which computational substrate is appropriate for each task.

Neural Inference Engines

GPUs, NPUs, and future neural accelerators perform:

- language generation
- probabilistic reasoning
- pattern recognition
- multimodal inference
- creative synthesis

These engines become specialized computational resources rather than executive controllers.

Symbolic Reasoning Engines

Symbolic processors implemented through CPUs, ASICs, FPGAs, or future specialized hardware perform deterministic operations including:

- rule-based inference
- logical verification
- mathematical reasoning
- constraint satisfaction
- graph traversal
- expert systems

Rather than approximating these operations probabilistically, deterministic engines execute them directly whenever appropriate.

Additional Specialized Engines

The architecture naturally extends to additional computational resources, including:

- search systems
- optimization engines
- simulation engines
- planning algorithms
- scientific computing
- future specialized accelerators

The architecture is intentionally hardware-agnostic.

It is organized around computational specialization rather than specific technologies.

Knowledge Infrastructure

Persistent knowledge should not be viewed as another computational engine.

Instead, it forms shared cognitive infrastructure owned by the Executive Kernel.

Possible implementations include:

- semantic knowledge graphs

- structured key-value state
- relational databases
- vector indices where appropriate
- append-only episodic logs

The Executive Kernel determines how these structures are maintained and queried.

Computational engines consume information from them but do not own them.

Execution Dispatch

Not every task requires full orchestration.

Simple conversational requests may be dispatched directly to neural inference.

More complex tasks may require:

- retrieval
- symbolic verification
- optimization
- planning
- neural synthesis

The Executive Kernel dynamically assembles the appropriate execution pipeline for each request.

Translation and Trade-offs

Introducing deterministic executive control introduces new engineering challenges.

Translating human language into structured cognitive state is itself an imperfect process.

Rather than assuming perfect translation, the Executive Kernel evaluates confidence before modifying persistent state.

When confidence is insufficient, the system requests clarification before committing changes.

Likewise, front-loaded context compilation introduces modest initial overhead.

However, this overhead may reduce repeated inference cycles, unnecessary computation, and human correction effort across longer interactions.

These represent architectural hypotheses requiring empirical validation.

Architectural Comparison

Dimension	Monolithic LLM	Agentic Systems	Tiered Cognitive Architecture
Executive Control	Neural model	Neural planner	Executive Cognitive Kernel
Memory	Conversation transcript	RAG + transcripts	Structured cognitive state
Context	Replay history	Replay plus retrieval	Compiled execution context

Logic	Probabilistic	Probabilistic	Deterministic where appropriate
Computation	Primarily neural	Primarily neural	Specialized heterogeneous computation
System State	Ephemeral	Semi-persistent	Persistent and deterministic

Conclusion

The history of AI has largely emphasized building increasingly capable models.

This proposal explores a complementary direction: building increasingly capable cognitive systems around those models.

Rather than assuming every cognitive function belongs inside a language model, this architecture separates executive cognition from specialized computation.

Language models remain indispensable for probabilistic inference and natural language generation.

They simply cease to function as the operating system of the AI.

Whether this architecture ultimately proves superior is an empirical question.

The purpose of this proposal is not to present a complete implementation, but to articulate an architectural principle for future exploration:

Intelligence emerges not from forcing every cognitive function through a single computational engine, but from coordinating specialized forms of

computation through a persistent executive layer that compiles structured cognitive state into task-specific execution contexts.

I think this version is the strongest yet. The biggest conceptual improvement is that it elevates **context compilation** from an implementation detail to the defining mechanism of the architecture. That makes the proposal less about "adding a memory system" and more about changing how AI systems organize cognition itself.